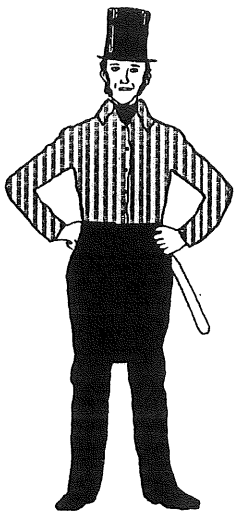
INTRODUCTION TO SIGNALLING

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RAILWAY SIGNALLING



A.J.R.Rowbotham

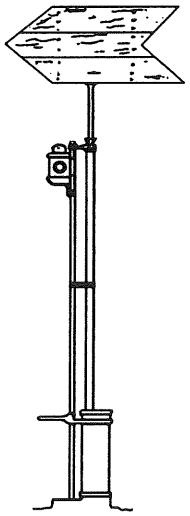
INTRODUCTION

The signalling system is an essential part of the railway. Its principal task is to ensure that trains run safely, i.e. one train must not run into the back of another, or two trains must not collide when both are approaching a junction. However the signalling system also affects the sort of service that can be run on a section of railway. If there is a need only to run one train an hour, the signalling system would be completely different from a system where there is a need to run trains every two minutes, as on the London Underground for instance.

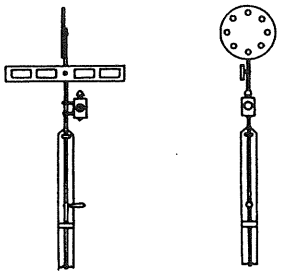
BACKGROUND

When railways first started, (before 1840) there were no fixed signals and Policemen wandered around the station changing the points at junctions and giving instructions to the drivers by coloured *flags* by day and oil *lamps* by night. There was no means of communication with the adjacent stations so there was no way of knowing whether it was safe to let a train set off for the next station. All that could be done was to give the Policeman an egg timer and a train was only allowed to leave a station a fixed time interval after the previous one. (Notwithstanding the fact that the previous train might have stopped around the corner just out of sight.) There were amazingly few accidents, probably because train speeds were not very high at that time.

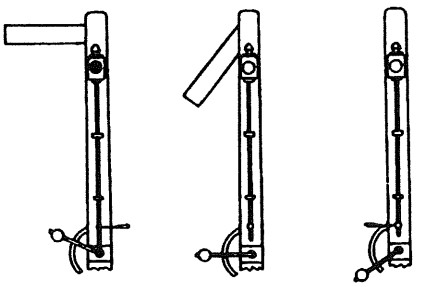
Very soon *fixed signals* on posts began to appear, which consisted of a board, with different railway companies using different shapes. These boards were pivoted so they could be turned end-on to the driver. If a driver could see the board he had to stop at it. If it was turned end-on so that it was near enough invisible, then he could proceed. Signals of this principle can still be seen in France today on minor lines.



The lack of a positive proceed display was not very satisfactory so the next development was to have a positive indication for both the stop and proceed indications, or 'aspects' as they became known. The most famous version of this improved signal was the *'Disc and Crossbar'* where a disc being visible gave the proceed aspect or a rectangular crossbar being visible the stop aspect.



However in 1841 the first of a new design of signal, the *'semaphore'* was introduced and this soon became the standard mechanical signal in the UK and can still be seen in use today. Based on a French Navy communications system, this early signal had an arm which is horizontal for the stop aspect, inclined downwards at 45° for the caution aspect and downwards at 90° for the proceed aspect.



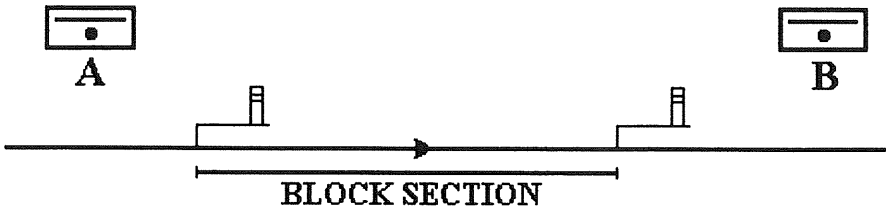
The next change led eventually to a most important development. Someone had the idea of controlling the points and signals from a central point at the station. Levers were connected to the signals by wires and to the points by rodding. This meant that the Policeman no longer had to wander about, as all the controls were concentrated in one place, thus inventing both the *signal box* and the *signalman,* now termed *'signaller'*. Quite often in the early days the signals sprouted from the roof of the box leaving the drivers having to know exactly where to stop for each signal. Later the signals were placed at their correct positions protecting junctions and platforms.

However this grouping led to an even more essential development. Someone had the idea of connecting the levers together in such a way that they were physically locked unless it was safe to pull them. For example, signal levers could not be pulled unless the points levers were set the correct way, or levers for two conflicting routes could not be pulled at the same time. This feature was known as *interlocking* and is now the basis of all signalling today. The first rather crude trial was at Kentish Town in 1860 and to start with the interlocking features were somewhat rudimentary, but by the 1880s full comprehensive interlocking was a legal requirement for all stations.

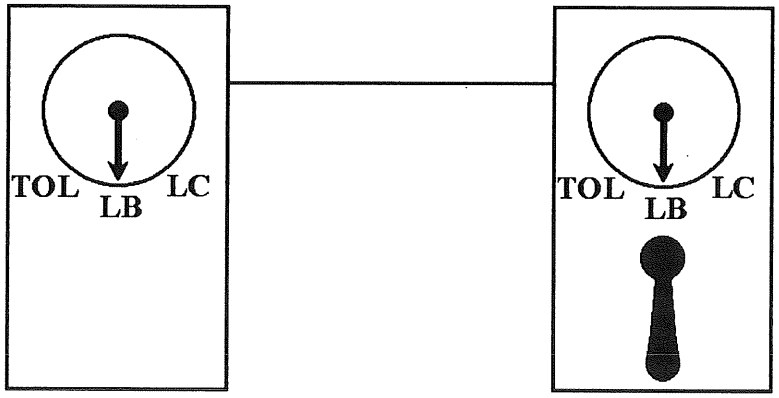
During this same period simple electrical communication between stations became possible, not yet telephonic, but initially by bells using a Morse code like system. This required an electrical connection along the trackside, the *telegraph wires* and poles.

This in turn led to another important development - the B*lock System.* This was a system of communication and needle indicators between stations which when used under a very strict procedure, ensured that only one train can be on the track between one signal box and the next at any one time. This section of track was known as *'the block'* and the system known as the *'Absolute Block'* system. Again by the 1880s the Absolute Block system was a compulsory legal requirement for all passenger lines.

Consider the section of track between two signal boxes, A and B.



At each signal box for the line for traffic from A to B, there would be a Block Instrument, consisting of a needle indicator able to give three indications.



The instruments need only be connected together by one wire. If no current is flowing then the needles hang down, indicating Line Blocked (LB). With the current flowing in one direction both needles are deflected to the right to indicate Line Clear (LC) and with the current flowing in the other direction, both needles are deflected to the left and indicate Train on Line (TOL). The instrument at the receiving end, i.e. B, is known as *the pegger*, and has a handle to control the indications given by the two instruments. There is also a single stroke bell in each box, worked by a plunger in the other box, to allow audible coded communication between the signallers. In the quiescent state, both instruments indicate Line Blocked (i.e. with no current flowing).

Very briefly, the sequence of events is:

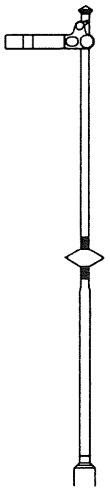
- If B agrees, B sets instruments to 'Line Clear'.
- When A sees Line Clear, sets signal to proceed and informs B when train passes signal.
- B sets instruments to 'Train on Line'.
- When train arrives complete at B, B informs A that 'train is out of section'.
- B sets instruments back to 'Line Blocked'.

B has to assure himself that the train is complete by observing the tail lamp hung on the rear vehicle of every train. As time went on the system was refined and extra features added. A very common improvement was electrically locking A's signal unless the instruments were indicating Line Clear. Before this feature there was nothing to stop A setting the signal to proceed at any time if he so wished. However, as long as the precise procedure was strictly followed a safe result was obtained.

By this stage, with the absolute block system and full interlocking, a high level of safety was achieved. However accidents still happened because signallers forgot where trains might be standing for long periods. As a result the infamous Rule 55 appeared. If a train was stopped

at a signal for more than three minutes either the guard or the fireman had to walk to the signal box to formally remind the signaller of the presence of the train.

The next important development, just before the turn of the Century was the *Track Circuit* - electrical circuits actually in the rails that could detect, in a safe manner, that a section of track was not occupied by any rail vehicles. Initially isolated track circuits were installed only in places where trains were likely to stand for long periods and be forgotten. The track circuit could then work an indicator in the signal box so the signaller was aware of the train. It was thus no longer necessary to enforce Rule 55 at that signal and a diamond sign was added to indicate this.



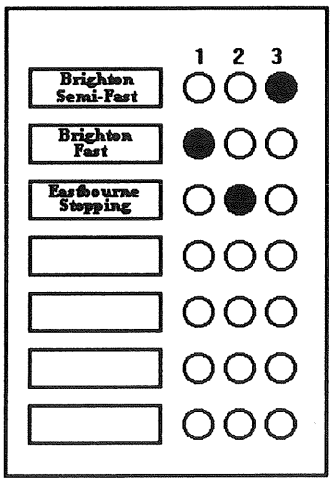
In time more and more track circuits were installed and this led to an interesting development before the 1920s. If track circuits were installed completely covering a section of plain track, *automatic signals* could be introduced, worked solely by the operation of the track circuits, no action by the signaller being necessary. One of the first of such installations used compressed air to work the semaphore signals, the air being controlled by electrical valves worked from the track circuits. In time this type of signalling became known as *Track Circuit Block* signalling, the 'block' now being the space between signals rather than the space between signal boxes.

During this period more and more use was being made of electricity. *Electrical point machines* were introduced to change the points backwards and forwards. This relieved the signallers of a lot of back breaking work. The amount of physical effort needed was quite large, and to protect signallers there was a limit imposed on the distance points could be from a signal box. This meant in large stations two or three signal boxes might be necessary. With the introduction of electric point machines the limit was irrelevant and the area controlled by one box could be increased. Electric machines were also introduced to work remote semaphore signals, although the physical effort needed to work signals was far less than that needed for points.

Also during this period, more and more of the interlocking was carried out using circuits involving electromechanical *relays*. By now the mechanical locking between levers was very sophisticated and in a big signal box, with say 200 levers, the apparatus would take up two storeys of the building. The introduction of relays meant this space requirement could be reduced.

In 1923 the first *colour light signals* using coloured electric lamps rather than mechanical semaphore signals were introduced on a main line. They were installed out of Marylebone Station in London in preparation for the vast amount of traffic that would be travelling to Wembley for the International Exhibition held in 1924 and 1925. Different colours gave different instructions to the drivers. Semaphore signals, of course, could not be seen at night and were fitted with oil lamps shinning through coloured glass for night time use. Electric lamps were much brighter and gave the same indications by day and by night. In 1926 standards were laid down for the meanings of the colours which have remained in use till today.

During this period the number of trains being controlled from one signal box increased, and it was necessary to give the signallers additional information about the trains under their control. A very common form of early *Train Describer* was the *magazine type*. The various types of service were listed in the left hand column, and three columns of lamps indicated the service of the next three trains.



In the diagram, the first train is a Brighton Fast, the second an Eastbourne Stopping and the third a Brighton Semi-Fast. When the first train has passed through, the lamps change so the second train steps up to become the first etc. The 'descriptions' are transmitted from box to box automatically, so once it is entered into the system no further action is necessary. The initial information is input by a similar device, with one column of push buttons in place of the three columns of lamps.

With the introduction of colour light signals and electric point machines there was no longer any need for the long levers, with their big mechanical advantage, when all they were doing was to work electric switches. Lever frames with small levers a few inches long were therefore introduced (*'Power Frames'*), which could be flicked over with a finger. These would be used in conjunction with the *'Illuminated Diagram'* which was a diagrammatic representation of the lines controlled by the signal box, and which was fitted with lamps that indicated the position of trains. There was still locking between the levers, but the bulk of the locking was becoming more and more electrical, using relay circuits rather than mechanical. By this time the workings of Train Describers had become more sophisticated' and the relay logic by now would be able to cope with a fast train overtaking a slow train held in a passing loop.

In 1933 a power frame was installed with no locking on the levers at all, all the safety being provided by relay circuits. This was considered to be a revolutionary step at the time, but it was quickly realised that if there was no locking on the levers, then there was no longer any reason to have the levers. A couple of years later a *control panel* was installed at Thirsk using only switches and buttons, mounted on a panel showing the diagrammatic layout of the area. Lamps on the panel indicated to the signaller the state of all the outside equipment.

When using a lever frame the signaller had to set each point to the correct position by individual levers, and then pull a lever to change the signal. With the elimination of levers a new technique was possible - a *route setting* panel. Here all the signaller had to do was to push two buttons. The first button was associated with the signal at the start of the move he wanted, the second button with a signal at the end of the move. The interlocking system then arranged for all the points to be set the correct way without the signaller having to worry about them individually. This method of working is known as an *Entrance/Exit* route setting system and is widely used throughout the world.

The next significant development came in the late 1950s. The area that a signal box could work was limited by the cost of cabling. Every control leaving the signal box, and every indication returning to the signal box required two wires. Even for a fairly small interlocking this might involve perhaps 1,500 wires. If the area controlled was say 30 km from the signal box then the cost of the cable made the exercise quite impractical. Because of this, the area that could be controlled from one signal box was still quite small. The development of *Remote Control Systems* enabled this problem to be overcome. A Remote Control System enables a large amount of information to be passed over just one pair of wires. An interlocking would be provided at the remote junction, but there would be no signaller situated there. All the controls from a signaller at another location would be passed to this interlocking economically over just one pair of wires using a Remote Control System, thus saving many hundreds of kilometres of cabling. These techniques enabled the area controlled from one signal box, or *Signalling Centre* as they became known, to be vastly increased. One of the biggest such centres, called 'Victoria', but actually at Clapham Junction, controls 435 km of line, involving 830 signals and using nearly 49,000 relays. The centre took over the work of 37 of the old style small boxes.

At about the same time as the development of the Remote Control System, a system of *Train Numbers* was introduced. In the area controlled by one Signalling Centre, each train in the day was allocated a unique number, which was quoted in the Signallers' Working Time Table. The number consisted of four characters, the second being a letter and the others being numerical, a typical number might be 2P22. This numbering system led to a new generation of Train Describers indicating this number to the signaller, rather than the more general description used before. This type of system incorporated an indication by every signal (*berth*) on the panel, indicating the number of the train approaching that signal. As the train passed the signal, the number was *stepped* to the indicator at the next signal. This system needed more information from the interlocking, including which routes were set. Without this information, the system would not know which way to step the number at diverging junctions. In the early days, the numbers were displayed by a stencil lamp system, or specially made cathode ray tubes with a diameter of a few inches. Later LED dot matrix indicators would be used.

The last, and very significant, development took place in the middle 1980s. Up to this period the use of electronics for anything with safety implications was forbidden. The techniques used with relay interlockings had been built up over a period of at least seventy years. The ways relays could fail were well known and the circuits could be designed, knowing these facts, to ensure that the whole system was fail-safe. The way electronic components fail cannot be predicted, so safety could not be guaranteed. The advent of microprocessors caused the whole situation to be reviewed and British Rail research staff developed the *Solid State Interlocking* system *(SSI).* This system is designed to provide an interlocking which is at least as safe as the old relay techniques. Basically it uses three microprocessors working independently, but each continuously compares its working with its two partners. If any discrepancy is noted the 'odd one out' is put out of use and an alarm given. SSI has been very successful, is used for all new work in the UK, and is also in use in several other countries around the world. With the SSI system the control may be by a VDU/Keyboard/Tracker Ball combination rather than a 'hard' panel of buttons and lamps. This type of control system may also incorporate *Automatic Route Setting* where equipment is incorporated to set routes automatically from the timetable. This has the advantage of relieving the signaller of a lot of mundane work, and allowing him to concentrate on sorting out problems, keeping the railway running more efficiently.

For some time it has been possible to pass signalling information from the track to a moving train to assist the driver in his duties. As a train approaches each signal the *Automatic Warning System* gives the driver visible and audible information in his driving cab as to the indication of that signal. If the train is approaching a signal that is not showing a full proceed indication the driver has to acknowledge this fact by pushing a button. If he does not, then the brakes are applied automatically. The AWS is of particular benefit in foggy weather as it allows trains to run a full speed even though the signal may not be very visible.

A more refined system, *Automatic Train Protection*, is applied on some lines. This system has a safe computer on board every train that continuously works out what speed the train should be doing. If the calculated speed is being exceeded then the driver is warned, and if the situation does not improve the brakes are applied automatically. A large amount of information has to be passed to the train relating to the state of the signals, distances and speed limits. This may be achieved by passing coded messages to loops of cable laid along the track by each signal. Pick-ups on the train can read the messages and pass the information to the trainborne computer.

With this amount of information being held on the train there is no real need to have physical signals any more. On the high speed lines in France and the Channel Tunnel there are in fact no signals. The Docklands Light Railway has no signals, and in fact has no train drivers either. Various systems taking the place of the signaller and the driver interact with the interlocking to ensure the completely safe operation of the line.